Last Updated: November 3rd, 2026

Hackathon - Data Analysis | Generative AI

📝 Instructions

For the 1st Hackathon, the goal is to practice the technologies you have learned in **Data Analysis with Python**.

Your project must tell a **story with data**: start from a question, get real data, clean it, explore it, visualize it, and finish with clear insights.

You may choose one of the following pathways.

Core Toolset (mandatory for everyone)

Whatever topic you choose, you **must** use:

- **Pandas / NumPy** — loading, cleaning and transforming data
- **Data cleaning** — missing values, duplicates, types, outliers
- **EDA (Exploratory Data Analysis)** — descriptive statistics, grouping, aggregations
- **Visualization** — Matplotlib, Seaborn or Plotly (minimum 4 meaningful charts)
- **Data sources** — CSV / Excel / JSON files **and** at least one external source (public API, SQL database, or web scraping)
- **Jupyter Notebook** — clean, commented, readable, executed from top to bottom
- **Insights** — a written conclusion: what did you find, and what would you recommend?

Pathway — Analysis & Storytelling

Focus on the depth of the analysis:

- Clear problem statement and hypotheses
- Advanced Pandas: `groupby`, `merge`, `pivot_table`, `apply`, time series
- Correlations and comparisons between segments
- A final notebook that reads like a report (titles, explanations, conclusions)
- Optional: statistical tests (scipy), or Faker to generate complementary dummy data

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💡 Ideas of General Topics for the Hackathon

Be creative! Here are ideas aligned with the toolset:

1. Open Data Explorer (API-based)

Build an analysis around a public API:

- World Bank, OpenWeather, REST Countries, Spotify, TMDB, football stats…
- Pull the data with `requests`, store it as JSON/CSV
- Compare countries, years, artists, teams…
- Deliver a notebook or a dashboard with filters

2. Sales & E-commerce Performance Analysis

Features:

- Revenue by product, category, region and month
- Top / flop products, seasonality, average basket
- Customer segmentation (RFM or simple clustering)
- Recommendations for the business

3. Personal Finance / Budget Analysis

Features:

- Categorize expenses and incomes over time

- Monthly trends, biggest spending categories, savings rate
- Currency conversion through an API
- Dashboard with a month selector

4. Health, Sport & Lifestyle Data

Examples:

- Sleep, steps, nutrition, training or public health datasets
- Correlations between habits and results
- Before/after comparisons, group comparisons
- Clear visual storytelling on what really matters

5. Text & Sentiment Analysis

Use:

- Reviews, tweets, news headlines or YouTube comments
- Cleaning text, word frequency, word cloud
- Sentiment scoring (TextBlob, VADER or an API)
- Evolution of sentiment over time or by product

6. Climate & Weather Trends Dashboard

Features:

- Fetch historical and real-time weather from an API
- Compare cities, seasons or decades
- Detect trends and anomalies
- Save favorite cities and build an interactive dashboard

📦 What you must deliver

- A **public GitHub repository** containing:
- Your notebook(s) and/or app code
- The dataset (or the script that fetches it)

- A **README** with: the problem statement, the data source, how to run the project, and your main findings
- A **2-minute video** of your screen explaining how your project works, the features and the tools used (you can use the Loom application)
- Both must be sent to the Slack channel `#hackathon` and written on the Google Doc

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🧠 Step of Innovation

DISCOVERY: Identify the focus, define the problem statement and the questions your data must answer
IDEATION: Generate ideas, find and validate your data sources
POC CREATION: Clean, explore and visualize — check feasibility of your analysis
EXECUTION: Finalize the insights and present your working prototype

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🏆 Judging Criteria

Teams will be judged on these four criteria, weighed equally. During judging, try to describe what you did for each criterion.

- **Data handling:** How well was the data collected, cleaned and prepared? Was the technical problem difficult?
- **Analysis & insights:** Are the conclusions supported by the data? Are the charts clear and honest?
- **Completion:** Does the project work end to end? Did the team achieve what they planned?
- **Learning:** Did the team stretch themselves? Did they try something new?

Judges are free to make decisions based on their gut feeling of which projects are the most impressive and most deserving.

These criteria do **not** include how well you pitch, how good the idea is, or how useful the project is in real life. Focus on building and learning.

⭐ Remember!

The competition is just one part of the hackathon.

Use this opportunity to **try something new**, **collaborate**, and **teach others**!

Hackathon #1

Progress 0% Complete

CODING DAILY DIGEST

COURSE NOTES

- Hackathon - Instructions — 5 XP
- Working with github — 5 XP
- Some advice for the hackathon — 5 XP
- ReadMe file - Github repo — 5 XP
- Mandatory Pitch — 5 XP
- 📺 Class Recordings — 5 XP

EXERCISES

QUIZ

ONLINE LEARNING

DAILY CHALLENGE

SURVEY

- Weekly feedback

WORKSHOP